

Anthony (Youwei) Zhen

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EDUCATION

Brown University

May 2027

Bachelor of Science in Applied Mathematics and Computer Science, GPA: 4.0

EXPERIENCE

SpaceX

Starbase, Texas

Starship Software Design Engineering Intern

June 2026 – September 2026

- Built and shipped a **design-review system** company-wide to a **5,000+ person engineering organization**; **165 engineers** logged **4,000+ feedback items** and **900+ comments**.
- Engineered an in-house **patch layer** over the **pdf.js engine**, extracting it from its static distribution and rewriting its frontend and backend in **TypeScript** to add **spatial annotation** on **PDF drawings** and **3D models** that captures coordinates to auto-generate screenshots pinpointing flagged components.
- Scaled **distributed job dispatch** to render **CAD drawings** on every design revision via concurrent worker queues.
- Identified that gating the **ECN** process on unaddressed drawing feedback was the missing lever, working with **10+ stakeholders** to drive a **10x company-wide increase in comment resolution**.

Refine.dev - YC 23

Remote

Machine Learning Engineer

March 2025 – May 2026

- Engineered production observability pipeline to detect **attention sinks** degradation in **long-context LLMs**
- Architected a distributed **code indexing system** using Pinecone and Redis caching, achieving **sub-100ms latency retrieval** across **150+ repositories** and reducing **LLM token consumption by 20%**.
- Optimized inference throughput to **2000+ token/second** by implementing **asynchronous API orchestration**, and **load-balancing** across multiple model providers.
- Designed and implemented **multi-agent system** with **cost-aware routing layer** and real-time prompt complexity classification, dynamically dispatching requests based on **semantic difficulty scoring** and **latency-cost tradeoff**.

Index Tax & Financial

Remote

Lead Software Engineer

December 2024 – Present

- Engineered **full-stack platform** for tax financial services firm with **Django REST**, **Nuxt.js**, **Electron.js**, and **Expo**, reducing CPA workflow time by **50%** through automated data aggregation and cross-referencing.
- Architected fully autonomous **RAG pipeline** with **Redis** caching, **Celery** worker distribution, and custom **vector database** to process **500GB+** of client tax documents concurrently.
- Built **CI/CD pipeline** with **Docker** for automated deployment, remote updates, and silent software distribution.
- Designed **concurrency control system** with **dynamic resource allocation** across **2 GPUs** and **24 cores**, preventing database deadlocks during real-time file ingestion and secured infrastructure via **reverse tunneling**.

PROJECTS & AWARDS

EdgeCart – AI Food Waste Prevention | *Python, OpenCV, ResNet, Gemini API*

November 2025

- 4x Winner @ HackPrinceton (Capital One, Knot API, Gemini, YC Runner-Up)**: Developed an edge-computing vision system to detect produce freshness using custom ResNet-50 models
- Implemented high-precision **blemish segmentation** using Google Gemini, improving detection accuracy by **19.5%**
- Implemented **intelligent discount matching** using **Knot API** and **xAI** to target customers, cutting waste by **180M tons annually**

TempoRoll – Real-Time EEG Music Therapy | *Python, PyTorch, Cerebras Systems*

September 2025

- 1st Place @ HackMIT (Cerebras Track)**: Engineered a real-time brainwave classification system using a hybrid CNN/ResNet architecture
- Optimized Wafer-Scale Engines inference, achieving **85% accuracy** on EEG streams with **sub 200ms latency**
- Integrated wireless **BLE** EEG communication and generated personalized music to guide brainwaves to baseline

USACO Gold Level (Platinum Division)

February 2023

11x Hackathon Winner (Harvard, MIT, Cornell, Princeton 4x, McGill, Brown 2x, Yconic ai) 2025 - Present

TECHNICAL SKILLS

Languages: Python (Advanced), C++, Java, TypeScript, JavaScript, SQL, R, C#

AI & Machine Learning: PyTorch, TensorFlow, vLLM, RAG, Vector Search, Transformers, BERT, CNNs, ResNet

Backend & Infrastructure: FastAPI, Django, Flask, Docker, Kubernetes, Nginx, PostgreSQL, Redis, MongoDB, CI/CD

Web Technologies: Next.js, React, Vue.js, Nuxt.js, Node.js, WebSockets, REST APIs

Spoken Languages: English, Chinese (Mandarin and Cantonese)